

QUBO Formulation: Single-Track Railway Dispatching Problem

Single-Track Railway Dispatching Problem

Problem Statement. Given a set of N trains $\mathcal{T} = \{0, \dots, N-1\}$ that must traverse a shared single-track segment, each train $t \in \mathcal{T}$ is associated with a set of M_t feasible timing options $\mathcal{O}_t = \{0, \dots, M_t-1\}$. Each option $o \in \mathcal{O}_t$ incurs a non-negative delay cost $c_{t,o}$. Certain pairs of options from different trains are mutually exclusive due to track occupancy conflicts, denoted by the conflict set $\mathcal{C} \subseteq \mathcal{T} \times \mathcal{O} \times \mathcal{T} \times \mathcal{O}$. The task is to select exactly one timing option for every train such that no two selected options conflict, thereby minimizing the total dispatch cost.

Decision Variables. For each train $t \in \mathcal{T}$ and each feasible option $o \in \mathcal{O}_t$, introduce a binary decision variable $x_{t,o} \in \{0, 1\}$. The variable $x_{t,o}$ equals 1 if train t is assigned timing option o , and 0 otherwise. The full decision vector is $\mathbf{x} = (x_{t,o})_{t \in \mathcal{T}, o \in \mathcal{O}_t} \in \{0, 1\}^n$, where $n = \sum_{t \in \mathcal{T}} M_t$.

QUBO Derivation. Step 1 — Original Objective. The problem requires minimizing the sum of selected delay costs plus a large penalty $P > 0$ for each selected conflicting pair:

$$\min_{\mathbf{x}} f(\mathbf{x}) = \sum_{t \in \mathcal{T}} \sum_{o \in \mathcal{O}_t} c_{t,o} x_{t,o} + P \sum_{((t_1, o_1), (t_2, o_2)) \in \mathcal{C}} x_{t_1, o_1} x_{t_2, o_2}. \quad (1)$$

Step 2 — Constraints. Each train must be assigned exactly one timing option:

$$\sum_{o \in \mathcal{O}_t} x_{t,o} = 1, \quad \forall t \in \mathcal{T}. \quad (2)$$

Step 3 — Penalty Term Expansion. We enforce the constraint by adding a quadratic penalty with weight $\lambda > 0$:

$$\mathcal{P}(\mathbf{x}) = \lambda \sum_{t \in \mathcal{T}} \left(\sum_{o \in \mathcal{O}_t} x_{t,o} - 1 \right)^2. \quad (3)$$

Expanding the square and applying the idempotent property $x_{t,o}^2 = x_{t,o}$ for binary variables yields:

$$\begin{aligned}
\left(\sum_{o \in \mathcal{O}_t} x_{t,o} - 1 \right)^2 &= \sum_{o \in \mathcal{O}_t} x_{t,o}^2 + \sum_{o \in \mathcal{O}_t} \sum_{\substack{o' \in \mathcal{O}_t \\ o' \neq o}} x_{t,o} x_{t,o'} - 2 \sum_{o \in \mathcal{O}_t} x_{t,o} + 1 \\
&= \sum_{o \in \mathcal{O}_t} x_{t,o} + \sum_{o \neq o'} x_{t,o} x_{t,o'} - 2 \sum_{o \in \mathcal{O}_t} x_{t,o} + 1 \\
&= - \sum_{o \in \mathcal{O}_t} x_{t,o} + \sum_{o \neq o'} x_{t,o} x_{t,o'} + 1.
\end{aligned} \tag{4}$$

Summing over all trains, the total penalty contribution to the Hamiltonian is:

$$\mathcal{P}(\mathbf{x}) = \lambda \sum_{t \in \mathcal{T}} \left(- \sum_{o \in \mathcal{O}_t} x_{t,o} + \sum_{o \neq o'} x_{t,o} x_{t,o'} + 1 \right). \tag{5}$$

Step 4 — Combined QUBO Hamiltonian. Adding the original objective and the penalty term gives the unconstrained QUBO Hamiltonian $H(\mathbf{x})$:

$$\begin{aligned}
H(\mathbf{x}) &= \sum_{t \in \mathcal{T}} \sum_{o \in \mathcal{O}_t} c_{t,o} x_{t,o} + P \sum_{((t_1, o_1), (t_2, o_2)) \in \mathcal{C}} x_{t_1, o_1} x_{t_2, o_2} \\
&\quad + \lambda \sum_{t \in \mathcal{T}} \left(- \sum_{o \in \mathcal{O}_t} x_{t,o} + \sum_{o \neq o'} x_{t,o} x_{t,o'} + 1 \right).
\end{aligned} \tag{6}$$

Step 5 — Q Matrix Entries and Offset. Collecting linear and quadratic terms, the Hamiltonian takes the standard form $H(\mathbf{x}) = \sum_{i \leq j} Q_{i,j} x_i x_j + c$. Mapping each pair (t, o) to a unique index $i \in \{0, \dots, n-1\}$, the matrix entries and constant offset are:

$$Q_{(t,o), (t',o')} = \begin{cases} c_{t,o} - \lambda & \text{if } t = t' \text{ and } o = o', \\ P/2 & \text{if } ((t, o), (t', o')) \in \mathcal{C} \text{ and } (t, o) \neq (t', o'), \\ \lambda & \text{if } t = t' \text{ and } o \neq o', \\ 0 & \text{otherwise,} \end{cases} \tag{7}$$

$$c = \lambda N. \tag{8}$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any violation of the exactly-one constraint is strictly more costly than the maximum possible reduction in the original objective. The maximum possible decrease in delay cost from selecting a suboptimal option is bounded by $\max_{t,o} c_{t,o}$. Therefore, selecting $\lambda > \max_{t,o} c_{t,o}$ guarantees that any infeasible assignment (where $\sum_o x_{t,o} \neq 1$ for some t) incurs a penalty of at least λ , which dominates any feasible objective value.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^n$ recovers the optimal dispatch schedule because (i) for any feasible assignment, all constraint terms vanish, reducing $H(\mathbf{x})$ to the original delay cost plus conflict penalties; (ii) for any infeasible assignment, at least one train is assigned zero or multiple options, incurring a penalty of at least λ . Since λ exceeds the maximum possible objective value of any feasible solution, every infeasible point has strictly higher energy than the best feasible point. Consequently, the global minimum of $H(\mathbf{x})$ must be feasible and, by construction, minimizes the original objective function.

Illustrative Example. Consider a concrete instance with $N = 2$ trains and $M_0 = M_1 = 2$ options per train. The variables are mapped to indices $0 \rightarrow (0,0)$, $1 \rightarrow (0,1)$, $2 \rightarrow (1,0)$, $3 \rightarrow (1,1)$. The conflict set is $\mathcal{C} = \{((0,0), (1,0)), ((0,1), (1,1))\}$. Using the general formulas above, the concrete objective terms are $c_{0,0}x_0 + c_{0,1}x_1 + c_{1,0}x_2 + c_{1,1}x_3 + Px_0x_2 + Px_1x_3$. The penalty terms for the exactly-one constraints expand to $-\lambda x_0 - \lambda x_1 + 2\lambda x_0x_1 - \lambda x_2 - \lambda x_3 + 2\lambda x_2x_3$, with an offset of 2λ . The resulting Hamiltonian is:

$$\begin{aligned} H(\mathbf{x}) = & (c_{0,0} - \lambda)x_0 + (c_{0,1} - \lambda)x_1 + (c_{1,0} - \lambda)x_2 + (c_{1,1} - \lambda)x_3 \\ & + Px_0x_2 + Px_1x_3 + 2\lambda x_0x_1 + 2\lambda x_2x_3 + 2\lambda. \end{aligned} \tag{9}$$

For a specific numerical instantiation, let $c_{0,0} = 1$, $c_{0,1} = 5$, $c_{1,0} = 3$, $c_{1,1} = 2$, $P = 100$, $\lambda = 10$. The feasible assignments are $(x_0, x_1, x_2, x_3) = (1, 0, 0, 1)$ with energy $1 + 2 + 2(10) = 23$, and $(0, 1, 1, 0)$ with energy $5 + 3 + 2(10) = 28$. The global minimum is $\mathbf{x}^* = (1, 0, 0, 1)$, corresponding to selecting option 0 for train 0 and option 1 for train 1, yielding a minimum feasible delay cost of 3.